



INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Mid-Spring Semester Examination 2022-23

Date of Examination:

Session: (FN/AN)

Duration: 3 hrs. Full Marks: 30

Subject No.: CH62006

Subject: Real-Time Intelligent Process Systems

Department/Center/School: Chemical Engineering

Specific charts, graph paper, log book etc., required: NO

Special Instructions (if any): Do not ask any questions during Exam, if in doubt, make suitable assumptions and proceed.

- Q-1
- (i) What do you understand by Off-line, On-line and Real-Time systems ?
 - (ii) Describe the Hierarchical Automation System detailing the Level-wise components.
 - (iii) Describe with an example, Manual Control, Level-1 based Control and Level-2 based Intelligent Control

$3 + 4 + 3 = 10$

Q-2. For a steam Jacketed continuous stirred heating tank,

- (i) Develop the dynamic mathematical model
- (ii) State all the appropriate assumptions
- (iii) Draw the Information Flow Diagram (IFD) clearly explaining the role of Tear Variables
- (iv) Describe the implementation of the above model for a Real-Time application

$2 + 1 + 4 + 3 = 10$

Q-3. Describe a Fuzzy Controller with 2 Input and 1 Output variable and show clearly the Calculation steps for the following

- (i) Membership Functions for all the Variables
- (ii) Fuzzification of the crisp input values
- (iii) Fuzzy inference using the Rule Book
- (iv) De-fuzzification (using CG method and Height method) of the Fuzzy output into crisp output values

$2 + 2 + 3 + 3 = 10$



INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

End-Spring Semester Examination 2022-23

Date of Examination: 19/04/2023 Session: (FN/AN) Duration: 3 hrs.

Full Marks: 50

Subject No.: CH62006 Subject: Real-Time Intelligent Process Systems

Department/Center/School: Chemical Engineering

1. (i) What do you understand by Off-line, On-line and Real-Time systems ?
(ii) Describe the Hierarchical Automation System detailing the Level-wise components.
(iii) Explain with an example, Manual Control, Level-1 based Control and Level-2 based Intelligent Control. [3 + 4 + 3 = 10]
2. With examples, show the implementation of (i) GA assisted Fuzzy membership function optimization (ii) GA assisted ANN weights optimization. [5 + 5 = 10]
3. Solve the following non-linear constrained optimization problem using GA: //
Minimise $(x - 1)^2 + (y - 4.5)^2$
Subject to : $4.5x^2 + y - 1.8 < 0$; $2x - y - 1 > 0$; $0 < x$; $y > 4$
Show calculation (with accuracy of 1 decimal place) up to two generations.
Use crossover probability as 50% and mutation rate of 25%.
Clearly state all the steps. [10]
4. What do you mean by "Membership function" and "MIN-MAX" process of rule evaluation? A washing machine manufacturer wishes to develop a Fuzzy Logic based control system to determine the Rinse time depending on the Feed cloth weight and Dirt level. Propose such a system with sample calculations clearly highlighting the membership functions, fuzzification, rule evaluation and defuzzification processes. [10]
5. Using "error back propagation" algorithm, show how the weights are adjusted for a 2 Input - 3 Hidden - 1 Output - node ANN system used to train an XOR truth table. Show one set of complete calculations. Use sigmoidal function as the activation function in the nodes of Hidden and Output layer. What are "learning rate" and "momentum factor"? [8 + 2 = 10]
6. A given system generates the control action (Y) based on the Input (X) as follows:
If $(-10 < X < -5)$ Then $Y = 0.1 X + 7.0$
If $(-5 < X < +5)$ Then $Y = -0.5 X + 4.0$
If $(+5 < X < +10)$ Then $Y = 0.7 X - 2.0$
Convert the above crisp logic to a Fuzzy rule set (Low/ Mod/ High) and smoothen the piecewise linear control curve. Show the numerical calculation and plot the free-hand graph and compare the output of Crisp and Fuzzy computation. What are the different types of fuzzy controller? [7 + 3 = 10]

fuzzy Intervention